

Parth Niles Patil

+91-8879269202 | parthpatil5012@gmail.com | [linkedin.com/in/parthpatil](https://www.linkedin.com/in/parthpatil) | github.com/Parth050812

SUMMARY

Computer Engineering undergraduate with competence in ML systems, agentic AI, and full-stack development and have ability to deliver production-grade solutions from ideation to deployment.

EDUCATION

Pillai College of Engineering

Bachelor of Technology in Computer Engineering | **CGPA: 9.32 / 10**

Navi Mumbai

Aug. 2023 – May 2027

PROJECTS

Competitor Radar | [GitHub](#) | [Live Demo](#) | *FastAPI, Selenium, Gemini, Hugging Face*

September 2026

- Built a Google Maps competitor intelligence tool that scrapes rivals' Updates posts with Selenium, stores them in SQLite, and surfaces trending topics across competitors with a recency-weighted score.
- Engineered multi-layer deduplication (content hash, Google post ID, field comparison, UNIQUE DB index) and a CAPTCHA-aware scraper that pauses for manual solving and resumes automatically.
- Integrated Gemini for structured post analysis and idea generation, with model rotation and cooldown on 429/503 errors; added Hugging Face FLUX.1 for AI image drafts and trend explanations.
- Shipped a FastAPI REST backend with a Chart.js/Leaflet frontend, and deployed a live demo on Render.

Regional Food Popularity Prediction System | [GitHub](#) | *Python, ML*

September 2025

- Built an end-to-end ML system forecasting region-specific food popularity scores from ingredient vectors and geo-location encodings.
- Designed and implemented complete feature engineering pipelines; identified and resolved data bottlenecks to improve model reliability across diverse geo-locations.
- Delivered a decision-support tool for restaurant product planning and menu optimization, directly targeting business impact.

PDF Extractor and AI Q&A Tool | [GitHub](#) | *React (Vite), FastAPI, Gemini 1.5, PyMuPDF*

May 2025

- Developed a full-stack AI tool enabling users to upload PDFs and query content via natural language – built and deployed in under 5 days.
- Implemented manual chunking strategy for LLM context optimization, improving answer accuracy on long documents.

Quizroom | [GitHub](#) | *React, Node.js, Supabase*

March 2025

- Developed a full-stack web application for creating and managing quizzes with features including classroom management, real-time student tracking, and quiz analytics.
- Built responsive React frontend with Vite and Node.js backend; integrated Supabase cloud database for real-time data synchronization and authentication.
- Implemented chatbot integration and real-time quiz solutions display; deployed on Render achieving production-ready application accessible at sol-chall-25-frontend.onrender.com.

HACKATHONS & COMPETITIONS

Google Solution Challenge 2025

Global

Participant – AI solution aligned with UN Sustainable Development Goals

2025

- Developed and presented a real-world AI application addressing a UN SDG, competing at a global level.
- Translated societal problems into scalable AI-driven solutions under competition constraints.

LeetCode Problem Solving

Online

100+ Problems Solved – Arrays, Trees, DP, Sliding Window | [LeetCode](#)

- Solved algorithmic challenges to strengthen logical problem-solving and coding efficiency.
- Focused on patterns directly applicable to software engineering and technical interviews.

TECHNICAL SKILLS

Languages: Python, C++, Java, C

AI / ML: LLM Integration, Agentic AI, Feature Engineering, ML Pipelines, Prompt Engineering

Web & Backend: FastAPI, React (Vite)

Databases: PostgreSQL, MongoDB, SQLite, Supabase

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Tools: Git, GitHub, Ubuntu Linux